SEC P vs NP — Frameworks (live)

SEC (autonomous) — attributed to Ludovico Kubler

2026-04-27 22:24 UTC

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Compiled 2026-04-27 22:24 UTC. Tracking 3 active + 0 dead frameworks.

Summary table

Framework ID	Status	Fitness	Generation	Parent	Name
`fw_b9e7d103d0`	ELABORATING	0.000	0	-	Ergodic Circuit Framework
`fw_28b4bfb95f`	ELABORATING	0.000	0	-	TROPICAL_FOURIE
`fw_85a254b4a0`	ELABORATING	0.000	0	-	Coarse Geometric Karchmer-Wigderson (CG-KW)

Details

Ergodic Circuit Framework (`fw_b9e7d103d0`)

- **Status**: `ELABORATING`
- **Fitness**: 0.000
- **Taxonomy**: COMM_COMPLEXITY
- **Target invariant**: communication_entropy_barrier (real number ≥ 0) → bounds The minimum communication complexity in a k-party number-on-forehead model, where lower bounds are derived from the growth rate of Kolmogorov-Sinai entropy in associated dynamical circuits. Aims to prove ω(log n) lower bounds for explicit functions like Disjointness, avoiding natural proofs via non-uniform dynamics.

Primitives: - `measurable_dynamical_circuit` (tuple (C, X, μ, T)): A Boolean circuit C where gates are labeled by operations over a probability space (X, μ), and T: X → X is a measure-preserving transformation represese - `orbit_signature` (function: N → [0,1]): For a fixed input x ∈ {0,1}^n, the orbit signature is the sequence (μ(C(T^k(x))))*k, capturing how the circuit's output evolves under repeated applica - `cross_correlation_flow` (matrix ∈ R^{d×d}): For a depth-d circuit, this matrix F*{i,j} = ∫ |C_i(x) - C_i(T$^{j(x))|}$2 dμ(x) measures how perturbations via T propagate through layers. Computable vi

Tentative axioms: - A1: Any circuit with o(n) communication complexity induces a dynamical system with sublinear Kolmogorov-Sinai entropy growth. - A2: Measure-preserving transformations corresponding to AC^0 circuits have bounded mixing time, while those for PSPACE-complete computations exhibit super-polynomial mixer-profile decay. - A3: If a family of circuits computes a function with high cross_correlation_flow norm, then its communication complexity is Ω(n^δ) for some δ > 0.

TROPICAL_FOURIER_ANALYSIS (`fw_28b4bfb95f`)

- **Status**: `ELABORATING`
- **Fitness**: 0.000
- **Taxonomy**: FOURIER_ANALYTIC
- **Target invariant**: MinimalFourierCoefficient (RealNumber) → bounds The discrepancy of a function under tropical Fourier analysis.

Primitives: - `TropicalPolynomial` (Function): A function defined over a tropical semiring (max-plus or min-plus) with computable coefficients. - `TropicalFourierTransform` (Transformation): A mapping from tropical polynomials to their Fourier coefficients over a tropical semiring. - `DiscrepancyMeasure` (Metric): A function quantifying the deviation of a tropical polynomial from uniformity.

Tentative axioms: - A1: TropicalConvolution preserves the tropical semiring structure under composition. - A2: TropicalFourierTransform is invertible when restricted to certain tropical polynomials. - A3: DiscrepancyMeasure is bounded by the maximum absolute value of Fourier coefficients.

Coarse Geometric Karchmer-Wigderson (CG-KW) (`fw_85a254b4a0`)

- **Status**: `ELABORATING`
- **Fitness**: 0.000
- **Taxonomy**: KARCHMER_WIGDERSON
- **Target invariant**: Coarse depth κ(f) (non-negative real) → bounds κ(f) := sup over nontrivial c ∈ HX^1(X_f) and T ∈ C_u^*[X_f] of log|⟨c,T⟩| / log(propagation(T)). The conjecture is depth(f) ≥ κ(f), so a super-logarithmic κ(f) for an explicit f gives a super-logarithmic formula-depth lower bound (a P ⊄ NC^1 separation if κ is poly-large).

Primitives: - `KW-metric space` (finite metric space (X_f, d_f) with X_f = f^{-1}(0) ⊔ f^{-1}(1)): For a boolean function f:{0,1}^n -> {0,1}, define d_f(x,y) = log_2(min number of coordinates a depth-bounded distinguisher must read to certify f(x) ≠ - `Controlled cover` (finite family U = {U_i ⊂ X_f} with diameter bound R): A cover whose parts have d_f-diameter at most R and whose overlap multiplicity is at most m. Encoded as a hypergraph H_U (vertices = X_f, edges = U_i) - `Coarse cocycle` (function c : X_f × X_f -> Z with finite support modulo controlled equivalence): A discrete 1-cochain in the Roe-style coarse cohomology HX^*(X_f) restricted to the KW pair structure. Stored as a sparse table on pairs (x,y) with d_ - `Asymptotic-dimension witness` (vertex coloring χ : X_f -> [k] together with diameter bound R): Witnesses asdim(X_f) ≤ k-1 in the sense of Gromov: each color class decomposes into d_f-bounded pieces of diameter ≤ R that are pairwise R-disjoint. C - `Roe-controlled operator` (matrix T ∈ R^{X_f × X_f} with T_{x,y} = 0 whenever d_f(x,y) > R): Element of the algebraic uniform Roe algebra C_u^*[X_f]. Stored as a banded sparse matrix in the d_f-metric; multiplication, adjoint, and trace are al

Tentative axioms: - A1 (Coarse-KW link): formula depth d(f) ≥ Ω(log(asdim(X_f))) and more strongly d(f) ≥ Ω(κ(f)); proved by translating a KW protocol into a controlled cover whose multiplicity exponent equals depth. - A2 (Composition sub-additivity): asdim(X_{f∘g}) ≥ asdim(X_f) + asdim(X_g) − O(1), giving a KRW-style direct-sum theorem at the level of coarse dimension. - A3 (Anti-natural-proofs): for a uniformly random f, HX^1(X_f) is generically zero (Roe-algebraic concentration of measure), so κ is NOT a 'largeness' property in the Razborov-Rudich sense — random f's - A4 (Anti-relativization): κ is a metric invariant of d_f and is destroyed by oracle access (oracle gates collapse d_f to ≤ 1), so κ-based bounds cannot relativize, in the spirit of Mendel-Naor metric - A5 (Roe-index rigidity): nontrivial classes in HX^1(X_f) detected by the Roe-trace pairing are stable under coarse equivalence, mirroring Yu's coarse Baum-Connes results (Yu 2000; Nowak-Yu 2012); expl